NE763 Exam 2 — 100 — Trevor Lillywhite

1) Beneficial SFR safety features

- High margin to boiling: operating ~ $T_{hot} = 550°C$, $T_{boil} \approx 880°C$
- High thermal conductivity of Na: keeps fuel cooler (more overpower required to melt)
- Large thermal mass (large Na volume with high heat capacity) - plus negative coefficient of reactivity ⇒ safely shut itself down
- Can operate at/near atmospheric pressure: no catastrophic pipe rupture causing coolant to flash (though it will burn in oxygen)
- High Na electrical conductivity - allows magnetically driven pumps (hermetically sealed), decreasing risk of coolant leakage.

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2) We alloy uranium in metallic fuels because the addition of ~10 wt% Zr (works best, other alloy constituents are possible) reduces swelling by stabilizing the BCC gamma phase (isotropic swelling compared to anisotropic alpha phase (orthorhombic)). The gamma phase existing at lower temperatures and having a higher solidus improves safety while suppressing swelling.

Further, Zr reduces fission product migration and can form a protective outer rind if exposed to O, N, or C during manufacturing, reducing FCCI and protecting the cladding.

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3) Metallic fuels have a low smear density (~75%) to reduce FCMI. Metallic fuels, even with Zr-stabilized γ phases, can swell ~30% during the first ~2% burnup before the fission gas pores interconnect and breach the fuel surface. The low smear density accommodates most of this swelling, and the cladding can withstand the strain from the remaining FCMI. The gap is filled with sodium to conduct heat away efficiently.

The large plenum (~1-1.5x the fuel slug length) allows the fission product gases to escape the fuel slug once pores interconnect and breach the surface, without causing cladding overpressurization.

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4) Constituent redistribution is the process of fuel constituents changing concentrations at varying radial positions over time vs. the original as-fabricated uniform distribution.

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This occurs because chemical potential gradients and Soret diffusion cause different species to move up or down the thermal gradient or to different phases that form at different radial temperatures. The center is the hottest, and Zr migrates up the thermal gradient due to a very negative $Q^*$. The uranium moves down the thermal gradient due to a mildly positive $Q^*$ and to accommodate the Zr migration. This leaves three radial regions:

Inner (hottest): Zr-enriched, U-depleted
Intermediate: U-enriched, Zr-depleted
Outer (coolest): Mostly unaffected, but maybe slightly Zr enriched.

Pu does not migrate substantially under these thermodynamic conditions, so it stays relatively uniform, but high concentrations of Pu (e.g., > 19 wt%) can accelerate the kinetics of Zr and U redistribution.

This causes different phases to form. The inside is $\gamma$ (BCC) phase, which occurs at high T and can accommodate a lot of Zr.

The middle is $\beta$-phase (often mixed with $\alpha$ or $\gamma$).

The outside is primarily $\alpha$ (orthorhombic) with very little Zr solubility, so the $\delta$ phase is intermingled with most of the Zr.

This can affect swelling, as $\alpha$ phase is more rigid while $\gamma$ is more soft and porous. There is more radial than axial expansion due to a biaxial loading condition.

5) Thermal conductivity degrades with burnup as fission product gases form pores that don't conduct well. As more pores [redacted] interjoin, they eventually breach the surface and escape. Thermal conductivity is most degraded right before this happens. Fortunately, after the gases escape, sodium bond fills some of the pores (~15-20%) and restores much of the thermal conductivity, leading to a sudden temperature drop that later equilibrates.

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6) FCCI is fuel cladding chemical interaction. These chemical interactions can degrade fuel performance by causing cladding wastage (existing cladding failure / fission product release) or formation of low $T_{melt}$ eutectics in the fuel that can cause fuel to [redacted] melt and cause issues like mess with signals sent to neutronics instrumentation.

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In the fuel, insoluble fission products like rare earths and noble metals can migrate to the low T periphery. If they aren't restrained by the Zr rind, they can attack the cladding and cladding constituents can diffuse into the fuel.

Fuel species: Ce, Nd, and other lanthanides migrate into cladding and can form peritectic mixtures like $Fe_{17}Nd_2$ or $Fe_{17}Nd_5$ ← (one of the slides incorrectly says 6)

Cladding species: Ni (if abundant present, like in SS-316 or D9), and Fe migrate the most. Cr can migrate a bit but largely piles up near the interface on the cladding side.

Those cladding species can lead to compounds like $(U, Pu)Fe_2$ or $(U, Pu)_6Fe$.

$Pu_6Fe$ is particularly troublesome because of its eutectic melting temp of ~700°C.

7) In MOX Fuel, very steep temperature gradients exist because of the low thermal conductivity. This, plus higher fuel temperatures than in LWR Fuels, causes unique restructuring.

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In the inner regions where T > 1800°C, small spherical pores have a larger thermal gradient (low conductivity through pore), and the U compounds ($UO_2$, plus $UO_3$ if oxygen potential is high enough) evaporate from the hot side and condense on the cool side. This moves U species down the thermal gradient and pores/voids up the thermal gradient. Pores join as they move, forming a lenticular shape until the reach the center and merge into an annular central void. The restructured uranium oxide behind it is in highly ordered crystals (columnar grain growth region).

Outside of this, T<1800°C but still high enough for grains to merge into larger grains. This is the equiaxed grain growth region.

At the periphery, T isn't high enough for grain growth so this region is largely unaffected (as-sintered region)

8) Pu oxides in this environment aren't as volatile as U oxides, so the Pu doesn't migrate outward but thermally migrates inward to become concentrated near the central void, back filling the $UO_2$ lattice positions.

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(There can also be localized Pu agglomerates from in complete mixing during the manufacturing process, leading to localized high burnup and large pore formation).

Oxygen moves down the thermal gradient, likely due to a combination of solid diffusion and vapor diffusion when combined with impurities (e.g., CO, $CO_2$, $H_2O$). This leads to a higher O/M ratio near the periphery and lower near the central void.

9) JOG stands for Joint Oxyde Gaine in French, which translates to "Oxide-Cladding Joint." It's a fission product oxide phase that forms around the outer surface of the fuel next to the cladding. As volatile fission products like Cs, Te, and I migrate to the periphery, they can be influenced by the oxygen potential and other species that precipitated near the periphery. Cs, Mo, and O can combine to form compounds like $Cs_2MoO_4$, though others can form too, depending on the oxygen potential and what fission products are present.

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These layers have 3 main effects:

- Increase thermal conductivity (Oxide more conductive than He bond)
- Increase mechanical stress, restraining fuel swelling but also pushing outward on cladding to a degree.
- Increasing FCCI, since it serves as a launching point for volatile species like Cs, Te, and I to attack the cladding through the matrix or intergranularly, especially if O/M is high enough to support reactions that dissolve the cladding's protective $Cr_2O_3$ layer.

10) Na corrosion is commonly divided into two categories.

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Type I: Solubility Limit driven dissolution of surface constituents into the sodium. The degree differs based on several factors, including binary diffusion coupling rates (high for Ni and Mn, medium for Fe, low for Cr ← relatively speaking)

Type II: Chemical reaction driven. If species like dissolved oxygen reach a surface, they can react (e.g., forming complex double oxides with Cr or other constituents), increasing solubility and corrosion rates.

Thermal Gradient Mass Transfer occurs as solubility limits decrease at lower $T_i$, so hotter surfaces corrode more and the dissolved species are dumped in cooler areas.

Carburization/decarburization can occur based on temperature and carbon chemical potential differences between hot/ferritic and cold/austenitic surfaces.